APP DEVELOPMENT MASTER ROADMAP

A full 18-phase guide covering Flutter & Dart, Mobile Architecture, APIs, Security, Publishing & Practical Apps.

PHASE 1 App Development Fundamentals

1. Intro to App Development & App Types
2. Native vs Cross-Platform vs Hybrid Apps
3. Android & iOS System Architecture
4. Mobile OS & Client-Server Architecture
5. Mobile Application Lifecycle
6. Development Environments & UI/UX Basics

PHASE 2 Programming Fundamentals

1. Variables, Data Types & Operators
2. Conditionals, Loops & Functions
3. Arrays, Strings, Lists & Dictionaries
4. Object-Oriented Programming (OOP)
5. Classes, Objects, Inheritance & Polymorphism
6. Encapsulation, Exception & File Handling
7. Data Structures & Algorithm Basics

PHASE 3 Dart Programming

1. Dart Syntax, Installation & Variables
2. Data Types, Operators, Loops & Functions
3. Lists, Sets, Maps & Collections
4. Null Safety & OOP Concepts (Abstract/Mixins)
5. Asynchronous Programming (Futures & Streams)
6. Exception Handling & Generics

PHASE 4 Flutter Fundamentals

1. Flutter Architecture & Installation
2. Android Studio & VS Code Configuration
3. Flutter Project Structure & CLI
4. Stateless vs Stateful Widgets & Lifecycle
5. Material Design & Cupertino Design
6. Hot Reload, Hot Restart & Debugging

PHASE 5 Flutter UI Development

1. Widget Tree, Container, Row & Column
2. Stack, Expanded, Flexible & Padding
3. Text, Image, Icon, Button & Card Widgets
4. Lists, Grids, AppBar & Nav Drawers
5. Dialogs, Snackbars & Custom Widgets
6. Themes, Styling, Animations & Responsive UI

PHASE 6 Navigation & Routing

1. Navigation Fundamentals & Navigator
2. Named Routes & Route Management
3. Passing & Returning Data Between Screens
4. Navigation Stack & Deep Linking
5. GoRouter & Nested Navigation
6. Auth-Based & Bottom Nav Routing

PHASE 7 State Management

1. State Management Fundamentals
2. setState & InheritedWidget
3. Provider & Riverpod
4. Bloc & Cubit Patterns
5. GetX Architecture
6. ValueNotifier & ChangeNotifier
7. Global vs Local State Architecture

PHASE 8 API Integration

1. REST API, HTTP Methods & Status Codes
2. JSON Fundamentals & Serialization
3. HTTP & Dio Packages in Flutter
4. Response Handling & Error Management
5. API Auth, Pagination & WebSockets

PHASE 9 Database & Storage

1. SQLite & Local SQL Fundamentals
2. Hive Database & Isar Database
3. Shared Preferences & Secure Storage
4. Offline Data Management & CRUD
5. Data Sync & Migration Strategies
6. Cloud Database Integration

PHASE 10 Backend & Cloud Integration

1. Firebase Auth, Firestore & Realtime DB
2. Firebase Storage, Messaging & Analytics
3. Supabase Auth, DB & Cloud Storage
4. REST Backend & Cloud Functions
5. Push Notifications Implementation

PHASE 11 Authentication & User Management

1. Email, Phone & OTP Verification Auth
2. Google & Apple OAuth Sign-In
3. Password Reset & Email Verification
4. JWT, Session & RBAC Management
5. User Profile & Multi-Factor Auth (MFA)

PHASE 12 Device Features & Hardware

1. Camera, Gallery & Microphone Integration
2. Location, GPS & Maps Services
3. Bluetooth, NFC & Wi-Fi Connectivity
4. Sensors (Accelerometer, Gyroscope)
5. Biometric Auth (Fingerprint/Face ID)
6. Background Services & Permissions

PHASE 13 Mobile App Security

1. Android & iOS Security Architectures
2. Secure Storage, Encryption & SSL Pinning
3. Root & Jailbreak Detection Mechanisms
4. Code Obfuscation & Key Management
5. OWASP Mobile Top 10, MASVS & MASTG

PHASE 14 App Testing & Debugging

1. Unit, Widget, Integration & UI Testing
2. Performance, Network & Device Testing
3. Flutter DevTools, Memory & CPU Profiling
4. Crash Analysis & Error Monitoring
5. Automated Testing & Test Coverage

PHASE 15 Advanced App Development

1. Clean Architecture, MVVM & Dependency Inj
2. Background Isolates & Performance Opt
3. Offline-First & WebRTC Real-Time Apps
4. Payment Gateways & In-App Purchases
5. App Localization & Accessibility

PHASE 16 Deployment & Publishing

1. APK & Android App Bundle (AAB) Gen
2. Keystore Mgmt & Manifest Config
3. Google Play Console & App Store Connect
4. App Store Optimization (ASO) & CI/CD
5. Beta Testing & Production Distribution

PHASE 17 Development Tools & Ecosystem

Flutter **Dart** **Android Studio** **VS Code** **Android SDK** **Flutter DevTools** **Git & GitHub** **Postman** **Firebase** **Supabase** **SQLite** **Figma** **Gradle** **Fastlane** **Codemagic**

PHASE 18 Practical Hands-on Projects

1. Calculator Application
2. To-Do List Application
3. Notes Application
4. Weather Application
5. Expense Tracker
6. Authentication Application
7. Chat Application
8. News Application
9. E-Commerce Application
10. Food Delivery Application
11. Fitness Tracking Application
12. College Management Application
13. Learning Management Application
14. Inventory Management Application
15. QR Code Scanner Application
16. GPS Tracking Application
17. Bluetooth Communication App
18. IoT Device Controller Application
19. Cybersecurity Learning Application
20. Complete Full-Stack Mobile Application